

- 1 The photograph shows part of a field of Kenaf plants, *Hibiscus cannabinus*.



(Source: © Design Pics Inc/ Alamy Stock Photo)

Crops of Kenaf plants are grown throughout Asia.

Several products are extracted from these plants:

- oil can be extracted from the seeds and this oil can be used as a biofuel
- fibres can be extracted from the stems and used to make sustainable materials.

(a) Describe an experiment to measure the tensile strength of fibres that have been extracted from these plants.

(6)

[illegible]

(b) Cellulose is a major component of plant fibres.

Describe the structure of a cellulose molecule.

(3)

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(c) Give **one** reason why the use of biofuels can reduce the effects of climate change.

(1)

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(Total for Question 1 = 10 marks)